ANY FPPF COVERING IS AN FPQC COVERING

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span. The same-chapter a-fortiori input is not carded and is declared external.

Allowed standard prerequisites (editorial).

- Permanence block for flat and finitely presented morphisms. Morphisms of Spaces, lemmas composition-finite-presentation, composition-flat, etale-flat, etale-locally-finite-presentation, fppf-open (cross-chapter, named block). The compositions f_{ij} are flat, lfp and open.
- A-fortiori chain for the smaller topologies. lemma-zariski-etale-smooth-syntomic-fppf (same chapter, NOT carded, declared external). Gives the syntomic/smooth/etale/Zariski corollary.

Source reference. Original definition 03MQ: sources/stacks-spaces-topologies/pinned-original-source.tex, source lines 1467–1467. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0DFC: sources/stacks-spaces-topologies/pinned-original-source.tex, source lines 998–998. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03XQ: focused-seeds/geometry/S9-topo-zariski-etale-smooth-syntomic-fppf-fpqc/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03MN: focused-seeds/geometry/S9-topo-zariski-etale-smooth-syntomic-fppf-fpqc/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0467: focused-seeds/geometry/S9-topo-zariski-etale-smooth-syntomic-fppf-fpqc/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-topologies.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0468: focused-seeds/geometry/S9-topo-zariski-etale-smooth-syntomic-fppf-fpqc/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 042S: focused-seeds/geometry/S9-topo-zariski-etale-smooth-syntomic-fppf-fpqc/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Any fppf covering is an fpqc covering, and a fortiori, any syntomic, smooth, étale or Zariski covering is an fpqc covering.*

Proof. We will show that an fppf covering is an fpqc covering, and then the rest follows from Lemma 0DFC. Let $\{f_i : U_i \to U\}_{i \in I}$ be an fppf covering of algebraic spaces over S. By definition this means that the f_i are flat which checks the first condition of Definition 03MQ. To check the second, let $V \to U$ be a morphism with V affine. We may choose an étale covering $\{V_{ij} \to V \times_U U_i\}$ with V_{ij} affine. Then the compositions $f_{ij} : V_{ij} \to V \times_U U_i \to V$ are flat and locally of finite presentation as compositions of such (Morphisms of Spaces, Lemmas 03XQ, 03MN, 0467, and 0468). Hence these morphisms are open (Morphisms of Spaces, Lemma 042S) and we see that $|V| = \bigcup_{i \in I} \bigcup_{j \in J_i} f_{ij}(|V_{ij}|)$ is an open covering of $|V|$. Since $|V|$ is quasi-compact, this covering has a finite refinement. Say $V_{i_1 j_1}, \ldots, V_{i_N j_N}$ do the job. Then $\{V_{i_k j_k} \to V\}_{k=1,\ldots,N}$ is a standard fpqc covering of V refinining the family $\{U_i \times_U V \to V\}$. This finishes the proof. □